

CSE4067 - Introduction to Blockchain Programming

Fall 2024, Midterm

Student No:

1	2	3	4

Problem 1 (+15)

Suppose a person called Alice with bitcoin address A owns three unspent transaction outputs (i.e. UTXOs) holding 15, 5 and 4 bitcoins and makes a payment of 17 bitcoins to a person called Bob with bitcoin address B. Draw three different possible transactions that may facilitate this payment.

Problem 2 (+20)

The table below shows the gas consumption of a contract function, the gas price, and the price of 1 ether in dollars (USD). Calculate how many dollars (USD) is spent in order to execute this function.

Gas Consumption	Gas Price	1 Ether
500000	8 Gwei	2000 USD

Problem 3 (+25)

Suppose you are given a file that contains the following bits (given in hexadecimal):

237ba7528b68ccaed5bf

Divide the file into data chunks of 16 bits and construct the Merkle tree. Take the hash function to be the 8-bit block-wise XOR function.

Problem 4 (+40)

Write the following Solidity smart contract that will implement a voting service:

- Anyone that pays 1 ether can vote on a proposal with yes/no answer.
- Anyone that pays 20 ethers can post a proposal. The proposal is entered as (text of proposal, duration in days).
- The proposals are identified by an integer 1, 2, 3, ... in the order of their entry.
- Anyone can display the text and the deadline of the proposal by referring to the proposal number.
- Anyone can display the number of proposals.
- Anyone can vote for a proposal by referring to the proposal number.
- After the voting period ends, voters are reimbursed as follows:
(1 ether they paid) + (20/(no. of proposal voters)) ethers for the voted proposal
- A withdraw function can be called in order to withdraw reimbursed ethers.